Brandon Bailey

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Software Engineer

SUMMARY

Software engineer and Computer Science student at University of Washington with production experience building distributed applications, high-performance networking tools, and scalable full-stack platforms. Delivered end-to-end features in Go, Python, and TypeScript, including a secure L3 VPN and real-time collaborative web services used by over 100 active customers.

EDUCATION

B.S., Computer Science, Math Minor (3.91 GPA)
University of Washington, Paul Allen School of Computer Science
SEPT 2023 - DEC 2026 (expected) | Seattle, Washington

COURSEWORK – Distributed Systems · Datacenters · DS & Algorithms · Networking

LANGUAGES – Go · C/C++ · Java · TypeScript · Python

TOOLS – Linux · TCP/UDP/IP · NFTables · NetLink · TensorFlow · NumPy · Cilium · Kubernetes · GCP

EMPLOYMENT

HyperRep, LLC — *Software Engineer*

MAY 2025 - CURRENT | Remote

- Built and shipped production features across frontend and backend using TypeScript (Next.js), Python, and SQL, supporting over 100 commercial clients on a customer-facing platform.
- Participated in technical discussions and iteration cycles to improve system scalability and user experience based on internal feedback and usage patterns.
- Designed and implemented real-time collaborative document editing, ensuring correctness under concurrent access using yjs encoding and conflict-free synchronization techniques.
- Collaborated with teammates on API design, data modeling, and performance improvements, balancing reliability, maintainability, and feature velocity.

PERSONAL PROJECTS

Custom VPN System — *Systems & Network Developement* NOV 2025 - DEC 2025

- Built a secure Layer-3 VPN in Go with encrypted tunneling and packet forwarding, enabling reliable and high-performance network connectivity.
- Implemented a multi-threaded packet pipeline using water, netlink, and netfilter/iptables APIs, including a connection handshake to exchange encryption keys and manage client traffic securely.
- Validated system behavior under concurrent load and network edge cases, strengthening understanding of availability, reliability, and performance tradeoffs in networked systems.

NetSim + NewReno TCP — *Python Network Systems Design* MAR 2026 - MAY 2026

- Built a NewReno-TCP-like transport protocol and discrete-event network simulator in Python, implementing sliding-window flow control, adaptive retransmission timers, congestion control, and selective acknowledgments under deterministic and adversarial network conditions.
- Designed and validated protocol behavior across packet loss, delay, and reordering scenarios, strengthening understanding of reliability, fairness, and performance tradeoffs in distributed systems.

COURSE PROJECTS

Paxos Implementation — *Java & Distributed Systems*

Implemented Multi-Paxos in Java, including leader election, log replication, failure handling, and consensus, ensuring safety and liveness across distributed nodes under loss, reordering, and simulated network faults.

Software-Defined Networking Controller — *Network Systems*

Developed an SDN controller in Python using POX to manage a Mininet virtual network, implementing custom routing, flow-rule installation, event-driven switch handling, and OpenFlow pipelines for packet forwarding, topology discovery, and adaptive flow control.